EDGE COMPUTING INNOVATION

TRINITY

REAL-TIME CONSTRUCTION SITE SAFETY MONITORING SYSTEM BASED ON EDGE COMPUTING AND COMPUTER VISION

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1. Motivation

Vietnam recorded **7,004** occupational accidents and 658 fatalities in 2025. Human negligence remains a key factor, with **13.75%** of cases involving safety procedure violations and **6.25%** due to failure to use Personal Protective Equipment (PPE) [1]. Meanwhile traditional CCTV monitoring by security personnel lead to fatigue and missed short-duration violations. Cloud-based systems also face high latency (500–1000 ms), heavy bandwidth usage (4–8 Mbps per camera), and significant privacy risks due to continuous raw video transmission.

Related Works	Research Gaps
YOLO-based PPE detection with attention modules [2]	Object-focused detection without hazard context.
Edge computing for construction monitoring [3,4]	Bandwidth savings mainly reported as theoretical estimates.
Frame-by-frame vision monitoring [5]	High false-positive rate caused by transient detections.
Model compression for edge AI [6]	Limited validation in real-world deployment scenarios.

Table 1: Related works

Contributions

- ***Context-Aware Alerting:*** *Worker location with PPE status to generate zone-aware WARNING/CRITICAL alerts.*
- ***Bandwidth Efficiency:*** *Reduced upstream traffic by* ***99.93% (1.35 MB/hour).***
- ***Temporal Smoothing:*** *Five-frame confirmation reduced false alarms by* ***94.6%****, maintaining* ***167 ms*** *alert latency.*
- ***Deployment Benchmark:*** *Validated the ONNX FP32 deployment pipeline, achieving* ***131 FPS*** *in an edge-oriented GPU proxy environment.*

2. Objective

- Develop an Edge AI system for real-time construction site safety monitoring.
- Detect PPE violations and hazardous situations automatically.
- Provide low-latency on-site alerts.
- Reduce false alarms through temporal verification.
- Minimize bandwidth and preserve privacy using an edge-first architecture.

Figure 1: PPE violated construction site

3. Dataset

The **Ultralytics Construction-PPE** dataset was used as the base dataset. Two additional public datasets were merged through class remapping to increase training samples and improve class balance.

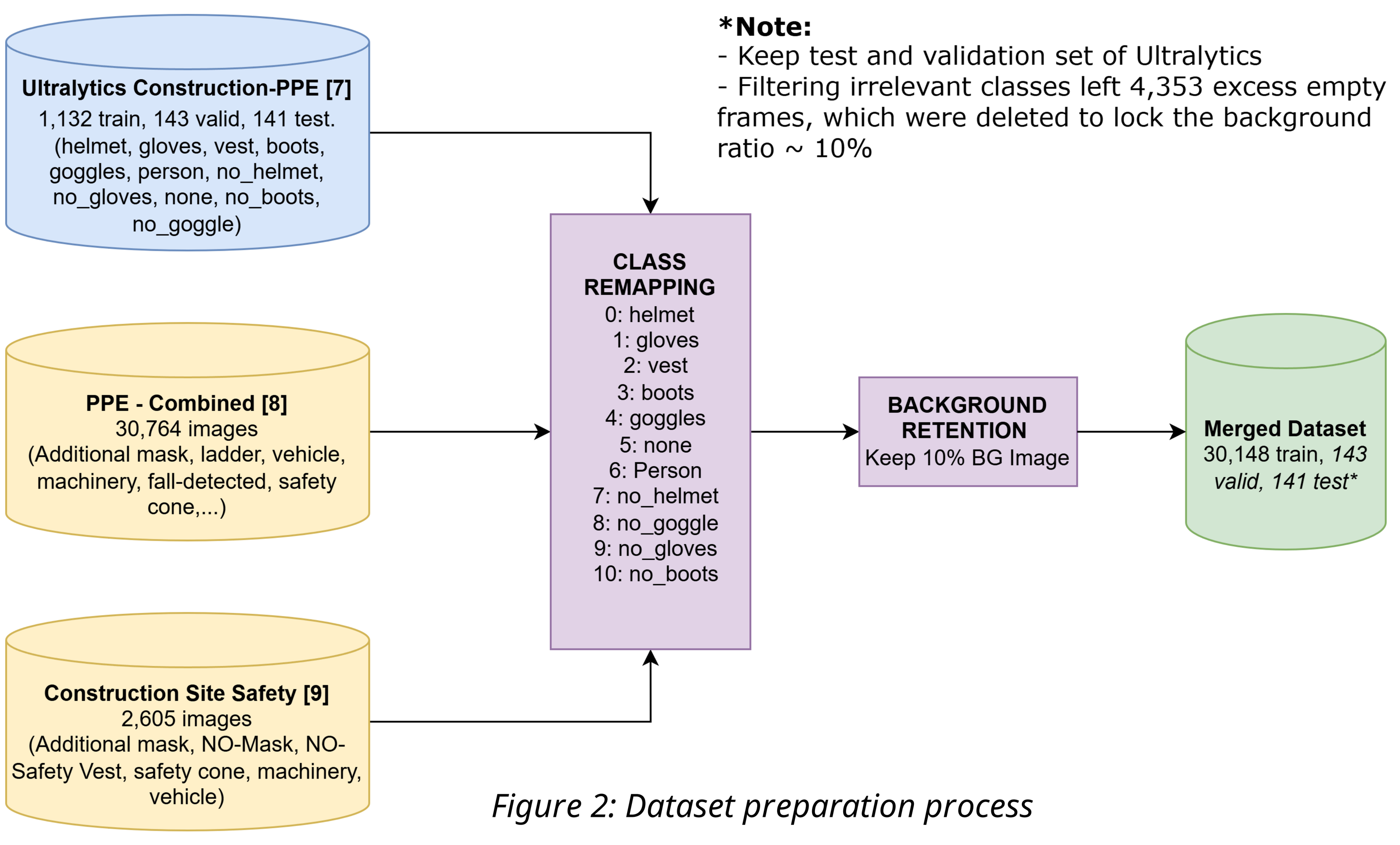

Figure 2: Dataset preparation process

b) Overall method

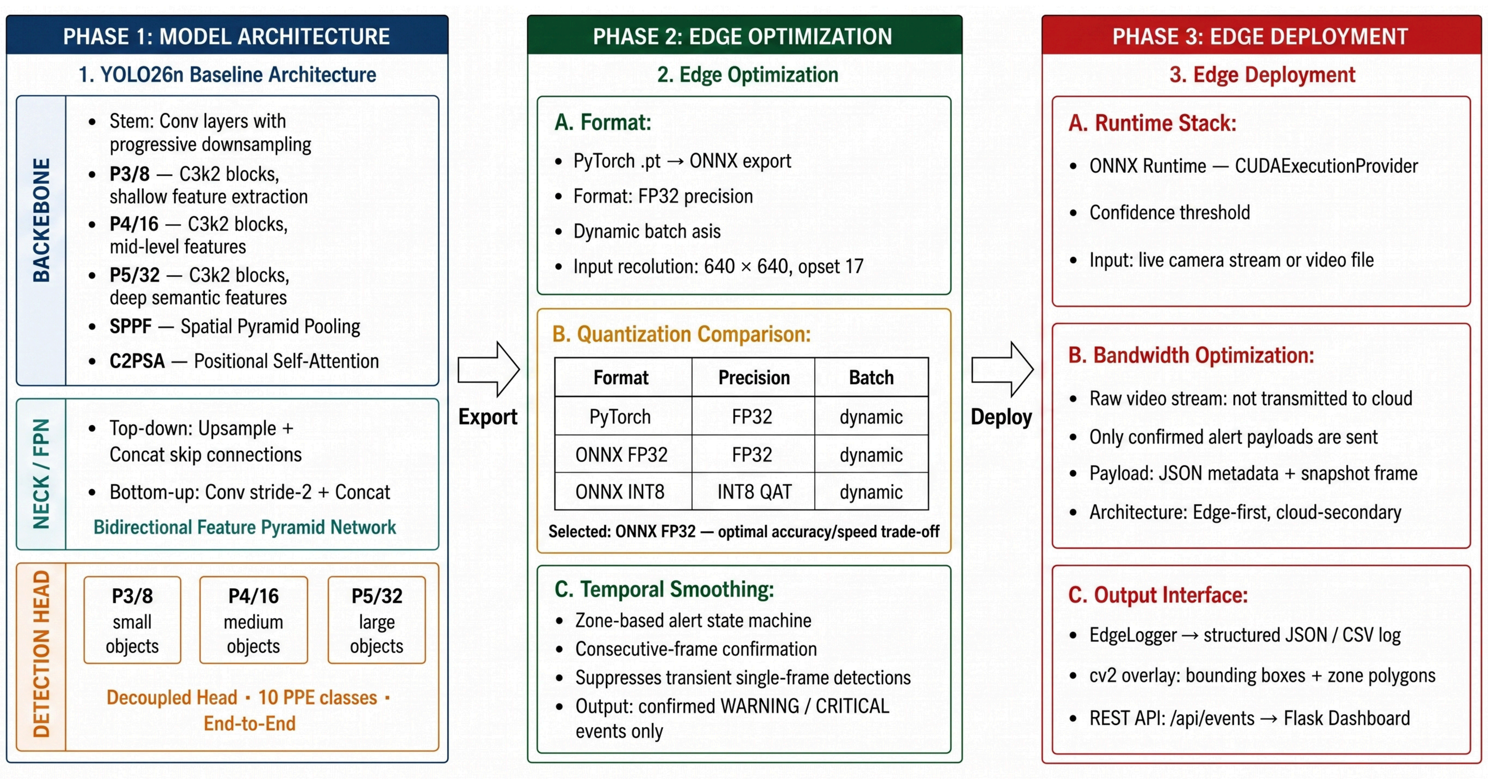

Figure 4: Overall method

4. Methodology

a) Workflow

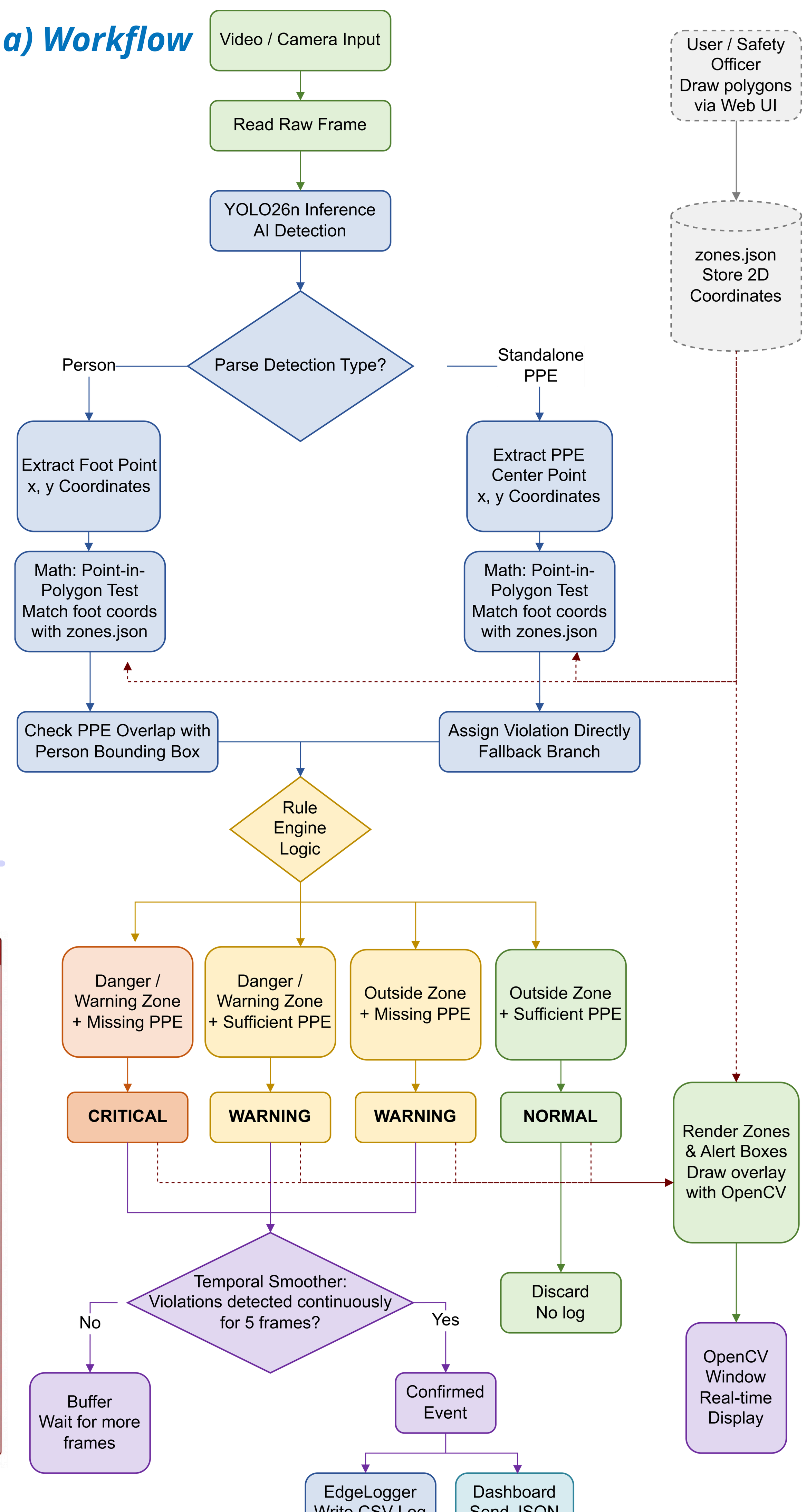

Figure 3: Workflow

5. Results

Model Format	Model Size	Latency (ms/img)	Throughput (FPS)	Overall mAP@0.5	Violation mAP@0.5
ONNX FP32 (Dynamic B=8)	10.3 MB	**7.61**	**131**	0.749	**0.671**
ONNX FP16 (Dynamic B=4)	**5.7 MB**	9.05	110	0.750	0.671

Table 2: ONNX Quantization Results

Optimization Target	Baseline State	Edge-Optimized Result
Alert Reliability (baseline_yolo26n ONNX FP32)	124 alerts (No temporal filter)	6 alerts (N = 5 frames)
Network Efficiency	1.897 GB/hr (Raw video)	1.35 MB/hr (Logs + Snapshots)*

**Note: Network efficiency assumes 1 event/min.* *Table 3: Edge Optimization Result*

Figure 5: Snapshot of a violated scene

mAP@0.5

Violated classes:
no_helmet **0.871**
no_goggle **0.877**
no_gloves **0.810**
(yolo26n_base_onnx_f32 B=8)

Edge-Optimized Efficiency

- Saved **99.93%** Bandwidth
- Reduced **94.6%** False positives
- Achieved **131 FPS** Edge throughput

End-to-End Pipeline

- **Context-Aware Logic:** Worker position + PPE status
- **Live Dashboard:** Real-time telemetry, snapshot logs, and zone mapping.

6. Conclusion

- Successfully deployed an Edge AI pipeline for real-time PPE monitoring, eliminating cloud-related latency, bandwidth, and privacy issues.
- The optimized baseline_yolo26n model (ONNX FP32) achieves 131 FPS at 2.93 ms/frame, enabling concurrent multi-camera processing.
- A 5-frame temporal smoothing algorithm reduces false alarms by 94.6%. The edge-first architecture reduces bandwidth consumption by 99.93% (1.35 MB/hour).
- ***Future Works:*** Enhance no_boots detection, integrate BIM for dynamic safety zones, and apply multi-modal sensor fusion for poor weather conditions.

7. References

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[2] Z. Wu et al., "Advancing construction safety: YOLOv8-CGS helmet detection model," PLOS One, vol. 20, no. 5, p. e0321713, May 2025.
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[9] Roboflow Universe Projects, "Construction Site Safety Dataset," Roboflow Universe, Jan. 2026.